Week 7 Field Guide — NLP in Pharma Commercial

Adverse events, field notes and social listening

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Monday, October 26, 2026 · Pairs with Week_07_NLP_Notebook.ipynb and the slides

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1. What this week is for

Most of what a commercial organization knows about its brand was typed, not tabulated: call notes, medical-information transcripts, forum posts, survey verbatims, advisory-board minutes. This week gives you one toolbox for all of it and one anchor use case where the stakes are highest — adverse-event (AE) detection — because an AE detector that misses a case is not a weaker model, it is an unreported case. Everything else (topics, sentiment, social listening, routing, search) reuses the same parts.

The one-sentence version

Define the target in the regulator's words → build the simplest detector that reads context → choose the threshold by cost → show the reviewer why → put rules and a human after the model → expect every new stream to behave differently.

2. The regulatory frame (what the detector must output)

Term	Meaning	Source
Adverse drug experience	Any adverse event associated with the use of a drug in humans, whether or not considered drug related.	21 CFR §314.80(a)
Serious	Death; life-threatening; hospitalization (initial or prolonged); persistent or significant disability; congenital anomaly; other medically important event.	21 CFR §314.80(a)
Unexpected	Not listed in the current labeling, or more severe or specific than listed.	21 CFR §314.80(a)
15-day Alert report	Serious and unexpected → report within 15 calendar days of initial receipt of the information.	21 CFR §314.80(c)(1)
The four elements	Identifiable patient; identifiable reporter; suspect drug; adverse experience. All four → a valid case.	FDA draft guidance 2001, §IV.B; ICH E2D
Company knowledge	FDA "will consider an applicant responsible for information known to its employees, affiliates, and contractors."	FDA draft guidance 2001, §X
Sponsored sites	"Applicants should review any Internet sites sponsored by them for adverse experience information." Sites not sponsored: no review duty; act if aware.	FDA draft guidance 2001, §VI.G
EU digital media	MAHs "should regularly screen internet or digital media under their management or responsibility."	EMA GVP Module VI Rev 2, VI.B.1.1.4
De-identification	Safe Harbor: 18 identifiers (names, dates, MRNs, …) removed before text counts as de-identified.	45 CFR §164.514(b)

The pipeline therefore answers four questions per record, in order: AE mentioned? → four elements present? → serious? → expected? — and computes a due date (awareness date + 15 days when serious and unexpected). Under-reporting is severe: a systematic review of 37 studies found a median under-reporting rate of 94% (Hazell & Shakir 2006). That is why mining the text is worth the trouble.

Insight versus obligation

The same detector feeds two very different consumers. Commercial analytics wants aggregates ("tolerability mentions rose after June") and can tolerate error. Pharmacovigilance needs individual cases with dates and cannot. Route to Safety first, insights second, and never let the analytics team be the only owner of an AE model.

3. The data (all synthetic)

File	Rows	What it is	Labels for students
glucoren_field_notes.csv	4,200	CRM call notes, Nov 2025–Oct 2026, 60 reps, 2,400 HCPs	pv_reviewed / pv_label_ae / pv_label_serious on 1,458 notes
patient_forum_posts.csv	3,000	Company-sponsored community + independent forum	pv_label_ae on 741 posts
medinfo_inquiries.csv	1,200	Contact-center inquiries, 8 intents	mi_team_intent / mi_team_ae_flag on 585
ae_lexicon.csv	77	Lay term → course preferred term (29 PTs), in_label, serious_default	—
faers_style_counts_glucoren.csv	26	Event counts for GLUCOREN vs all other drugs, FAERS-shaped	—
hcp_master_w7.csv	2,400	Specialty, region, segment per HCP	—

The "course PTs" are plain-English event names we made up for teaching. In practice events are coded in MedDRA, a licensed terminology maintained by the MSSO under ICH; your company's safety database uses it, and a real lexicon maps lay terms to MedDRA Lowest Level Terms.

4. Detector 1 — a lexicon

Flag a note if any of the 77 lay terms appears. On the reviewed notes: recall 79%, precision 42%, 270 flags. It fails in two families:

- Negation and general statements — "no reports of nausea", "asked about the nausea rate in the label". The word is present; nobody had anything. A lexicon has no notion of who or whether.
- Paraphrase, typos, abbreviations — "N/V", "GI issues", "sugar lows", "nuasea". The event is there; the string is not.

Keep the lexicon: it becomes the seriousness and expectedness lookup in §7. Stop using it as the detector.

5. Detector 2 — TF-IDF + logistic regression

Bag of words with 1–2-grams, TF-IDF weighting, L2-regularized logistic regression with class weights, trained on the reviewer's labels. One classic trick: negation scope marking — the three words after "no", "denied" or "without" get a NOT_ prefix (Pang, Lee & Vaithyanathan 2002), so "no nausea" and "nausea" are different features. Five-fold cross-validation: ROC-AUC 0.99, average precision 0.94. At the chosen threshold (0.47): recall 95%, precision 95%. On the 2,742 notes the reviewer never saw (instructor truth), recall 96% and precision 94%; the lexicon's recall there is 79%.

The threshold is a business decision

Threshold	Recall	Precision	Flags / week	Reviewer hours / week	AEs missed / year
0.20	97%	62%	13	1.3	12

0.30	97%	80%	10	1.0	14
0.47	95%	95%	8	0.8	20
0.50	93%	94%	8	0.8	29
0.70	79%	95%	7	0.7	89

Assumptions, stated: six minutes per note for a trained reviewer; about 84 notes a week; the reviewed sample is representative of the stream. In a safety setting the answer is almost always “high recall, and a human reads the flags”. Put the cost table in the model documentation — it is the justification an inspector will ask for.

Behavioural probes

Five sentences the model must get right on every retrain — a positive, its negation, a general statement, an abbreviated paraphrase, a serious case:

Probe	p(AE)
Patient developed nausea after starting GLUCOREN and stopped the drug.	0.94
Patient had no nausea after starting GLUCOREN; tolerating well.	0.34
HCP asked about the nausea rate in the label; no specific patient discussed.	0.07
One of his pts had N/V within 2 wks of starting, dose reduced.	0.99
A 61 y/o F was hospitalized with pancreatitis three weeks after starting GLUCOREN.	0.87

6. Opening the model — interpretability

> *If you cannot explain how your AI came to a decision, it does not belong in a compliance system.*
>
> — IQVIA, A New, AI-Augmented Horizon for Safety and Pharmacovigilance: Expert Predictions for 2026

Question	Scope	Method	Reference
What does the model think an AE looks like?	global	Coefficients of the linear model. Event words + case words on one side; negation and generality on the other.	Molnar 2022, ch. 5
Why was this note flagged?	local, exact	Word contribution = coefficient × TF-IDF weight. Shown as highlighted spans in the note.	—
Why — if the model were a black box?	local, model-agnostic	LIME: perturb the text (words on/off), query the model, weight neighbours by closeness, fit a local linear surrogate.	Ribeiro, Singh & Guestrin 2016
Same, with consistent credit	local + global	SHAP values (Shapley). Mean |SHAP| ranks by how often a term matters, not only how strongly.	Lundberg & Lee 2017
Which tokens, for a transformer?	local	Integrated Gradients. Attention weights are not reliable explanations by default.	Sundararajan et al. 2017; Jain & Wallace 2019; Wiegreffe & Pinter 2019

Rudin (2019) argues that for high-stakes decisions we should prefer models that are interpretable by construction over post-hoc explanations of black boxes. Tonight's detector is exactly that; LIME and SHAP are taught so that you can read them when a vendor's model is not. Three practical rules:

- Do the coefficient review before the PR curve. It is the cheapest model review you will ever run.
- Explanations are for finding problems. The leak check found payer names with large weights (“optumrx moved” +2.0, “cigna is” +1.8, “reports cigna” +1.8) — a shortcut from the training sample. The remedy is to mask entities (payer → “payer”) and re-fit, not to explain the shortcut away.
- Reviewers want the span, not the number. Build the explanation into the queue's user interface.

LIME by hand (the whole idea)

For one note: (1) make 400 variants with words switched off at random; (2) ask the model for p(AE) on each; (3) weight variants by how close they are to the original; (4) fit a weighted ridge regression from the on/off mask to p; (5) read its coefficients as word effects. Two cautions: explanations are local and can change with the sampling seed, and variants with words removed are sentences nobody writes, so the model is being asked about inputs off its data manifold.

7. From flag to case — rules after the model

Rule	How	Result on the stream
Identifiable patient	Regex: age/sex pattern, "one of his patients", a name, an MRN	458 flags → 418 valid cases
Seriousness	Lexicon terms: hospitalized, ER, ICU, died, pancreatitis, kidney injury, …	51 with serious terms
Unexpected	Lexicon column in_label = N	—
15-day candidate	valid ∧ serious ∧ unexpected; due = awareness + 15 days	50 candidates (51 true cases in the stream)
PHI in note	DOB, MRN, Mr./Mrs. + name, "patient First Last"	28 notes — scrub before storage or any external model

The queue runs at about 9 flags a week (0.9 reviewer-hours). Reps are trained to report AEs within one business day; the detector is a safety net on the CRM, not a replacement for the SOP — and when it surfaces a case the clock has been running since the call date. Off-label mentions (unsolicited questions about unapproved uses) are documented and routed to Medical Information; the rep must not engage. The same classifier family detects them — try it.

8. The rest of the toolbox

What the field hears — topic modeling

Non-negative matrix factorization (Lee & Seung 1999) on TF-IDF with a domain stop-word list, after concept normalization (every lexicon hit → one token). Topics are word clusters, not business themes: the model separates "copay card" from "prior auth" from "Medicare gap". Roll them up, name them yourself, and — where you already have a supervised detector — assign that theme with the detector (safety here). LDA (Blei, Ng & Jordan 2003) is the probabilistic classic; BERTopic (Grootendorst 2022) clusters sentence embeddings and usually gives cleaner themes on short notes.

Theme	Notes	Negative tone	AE-flag rate
Access / payer	945	37%	0%
Safety / tolerability	874	34%	52%
Samples / logistics	586	4%	0%
No-see / brief contact	508	24%	0%
Competitor	476	30%	0%
Dosing & combinations	423	21%	0%
Education / MSL	388	1%	0%

Safety share of notes: 18% before June 2026, 25% after — the trend the brand team sees here first.

Sentiment

A transparent word-list score (VADER-style, Hutto & Gilbert 2014) is good enough for a slope. When the decision depends on a level, train a classifier on your own labelled verbatims.

Social listening — domain shift

The notes-trained detector scored on forum posts: precision 64%, recall 91%. Trained on post labels (5-fold CV): precision 94%, recall 99%. A model is only as good as the distribution it was trained on; get labels on the new stream first. Social-media AEs skew to mild, symptom-level events and under-represent serious ones compared with regulatory databases (Golder, Norman & Loke 2015); Twitter event rankings correlated with FAERS, absolute counts did not (Freifeld et al. 2014).

Medical-information routing

An intent classifier (8 intents, accuracy 92%) routes inquiries to the MI specialist, Safety, Quality or Access. But 111 of 333 AE mentions sat inside non-AE intents (a dosing call that mentions nausea). Run the safety detector on every record, whatever its intent.

Signal detection

Once cases are in the safety database the question becomes: is this event reported with our drug more often than expected? Proportional Reporting Ratio PRR = (a/(a+b)) ÷ (c/(c+d)); Reporting Odds Ratio ROR = ad/bc with a 95% CI; Evans's screen PRR ≥ 2, χ^2 ≥ 4, n ≥ 3 (Evans, Waller & Davis 2001); Bayesian shrinkage (BCPNN, Bate et al. 1998) for sparse cells. Tonight's synthetic table flags Pancreatitis, Hypoglycaemia, Nausea; pancreatitis (PRR 3.6) is not in the label and goes to a signal-review committee. There is no denominator of exposed patients and under-reporting is heavy: a hypothesis, not a rate.

Semantic search

Latent Semantic Analysis (truncated SVD on TF-IDF; Deerwester et al. 1990) is the offline stand-in for embeddings. The swap-in is sentence-transformers (Reimers & Gurevych 2019): same code shape, better neighbours.

9. Domain models and the transformer route

BERT (Devlin et al. 2019) is pretrained on general English; BioBERT (Lee et al. 2020) continues pretraining on PubMed and PMC; ClinicalBERT (Alsentzer et al. 2019) on MIMIC clinical notes; PubMedBERT (Gu et al. 2021) pretrains from scratch on biomedical text and beats continued pretraining on the BLURB benchmark. Match the pretraining register to your text — rep notes and patient posts are general English with a few medical words; chart review and literature are not — and test the general model first: it is often within a point or two. The notebook's transformer cells are behind USE_TRANSFORMERS = False because class runs offline. At home:

Fine-tuning a classifier on the reviewed notes (sketch)

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification, Trainer, TrainingArguments
tok = AutoTokenizer.from_pretrained('dmis-lab/biobert-base-cased-v1.2')  # or 'bert-base-uncased'
model = AutoModelForSequenceClassification.from_pretrained('dmis-lab/biobert-base-cased-v1.2', num_labels=2)
enc = tok(list(rev.note_text), truncation=True, padding=True, max_length=96)  → a torch Dataset with labels y
Trainer(model, TrainingArguments('out', num_train_epochs=3, per_device_train_batch_size=16,
```

learning_rate=2e-5), train_dataset=…).train()

Evaluate on the same folds, at the same recall target, with the same probes — and with Integrated Gradients (captum) for the spans.

10. Where GenAI fits (preview)

Large language models change the economics of this pipeline, not its logic. Zero-shot extraction returns the four elements from a note with no training set; normalization to a coding dictionary and narrative drafting get cheap; retrieval + generation grounds answers in approved content. What does not change: the four elements, the seriousness criteria, the 15-day clock, human review of every case, PHI scrubbing before any text leaves the firewall, and the need to measure precision, recall and workload on your own labelled data. IQVIA reports that across industries more than 70% of GenAI experiments fail to reach production; the ones that survive are evaluated like the models above. The full session covers prompting and evaluation, hallucination, RAG over MLR-approved content, cost, and the FDA's January 2025 draft guidance on AI credibility for regulatory decisions.

11. Governance checklist

Item	Write down
Context of use	which stream, which decision, who acts, what happens on error (FDA 2025 credibility framework)
Data	source, period, reviewer labels and their noise, PHI handling
Performance	AUC/AP for comparison; the operating point with recall, precision, workload; held-out and behavioural tests
Explanations	global review, per-flag spans, leak checks
Human oversight	who reviews every flag; escalation; the SOP it sits beside — “a human-in-the-loop model … is the operating model itself” (IQVIA, Raizada 2025)
Monitoring	flag-rate drift, probes on every retrain, domain shift on new streams
Record	the PV System Master File (IQVIA governance paper; EMA GVP)

12. Vocabulary → source

Term	How tonight uses it	Source
Adverse experience; serious; unexpected; 15-day Alert report	The target the detector must output	U.S. Food and Drug Administration; International Council for Harmonisation 2003
The four minimum elements of a case	Identifiable patient, identifiable reporter, suspect drug, adverse experience	U.S. Food and Drug Administration 2001; International Council for Harmonisation 2003
Company knowledge; sponsored vs non-sponsored sites	A rep note is company knowledge; screen the media you control	U.S. Food and Drug Administration 2001; European Medicines Agency 2017
Under-reporting	Median 94% of ADRs never reported	Hazell 2006
De-identification (Safe Harbor)	Scrub names, dates, MRNs from free text	U.S. Department of Health and Human Services 2012
Lexicon → classical ML → transformers	The ladder of detectors	Karimi 2015; Devlin 2019; Lee 1240
Negation scope (NOT_ prefix)	Why "no nausea" is not nausea	Pang 2002
Precision–recall operating point	Recall first, reviewer workload second	Karimi 2015
Domain models vs general models	BioBERT, ClinicalBERT, PubMedBERT; when generic is enough	Lee 1240; Alsentzer 2019; Gu 2021
Embeddings; semantic search	Sentence-BERT; LSA as the offline stand-in	Reimers 2019; Deerwester 1990
Interpretability: global vs local;	Coefficients, LIME, SHAP, Integrated Gradients	Molnar 2022; Ribeiro 2016;

intrinsic vs post-hoc		Lundberg; Sundararajan 2017; Rudin 2019
Attention is (not) explanation	Caution on reading attention maps	Jain 2019; Wiegreffe 2019
Topic modeling; sentiment	NMF, LDA, BERTopic; VADER-style lexicons	Lee 1999; Blei 1022; Grootendorst 2022; Hutto 2014
Social-media pharmacovigilance; domain shift	What social listening finds and misses	Sarker 2015; Nikfarjam 2015; Golder 2015; Freifeld 2014; Weissenbacher 2019
Disproportionality: PRR, ROR, BCPNN	Signal detection on spontaneous reports	Evans 2001; Bate 1998; Harpaz 1021
AI governance in PV	Human in the loop; PSMF documentation; credibility framework	IQVIA 2025; IQVIA (Raizada) 2025; U.S. Food and Drug Administration 2025; FDA, Health Canada, MHRA 2021